

Status of the Nuclotron

A.Sidorin

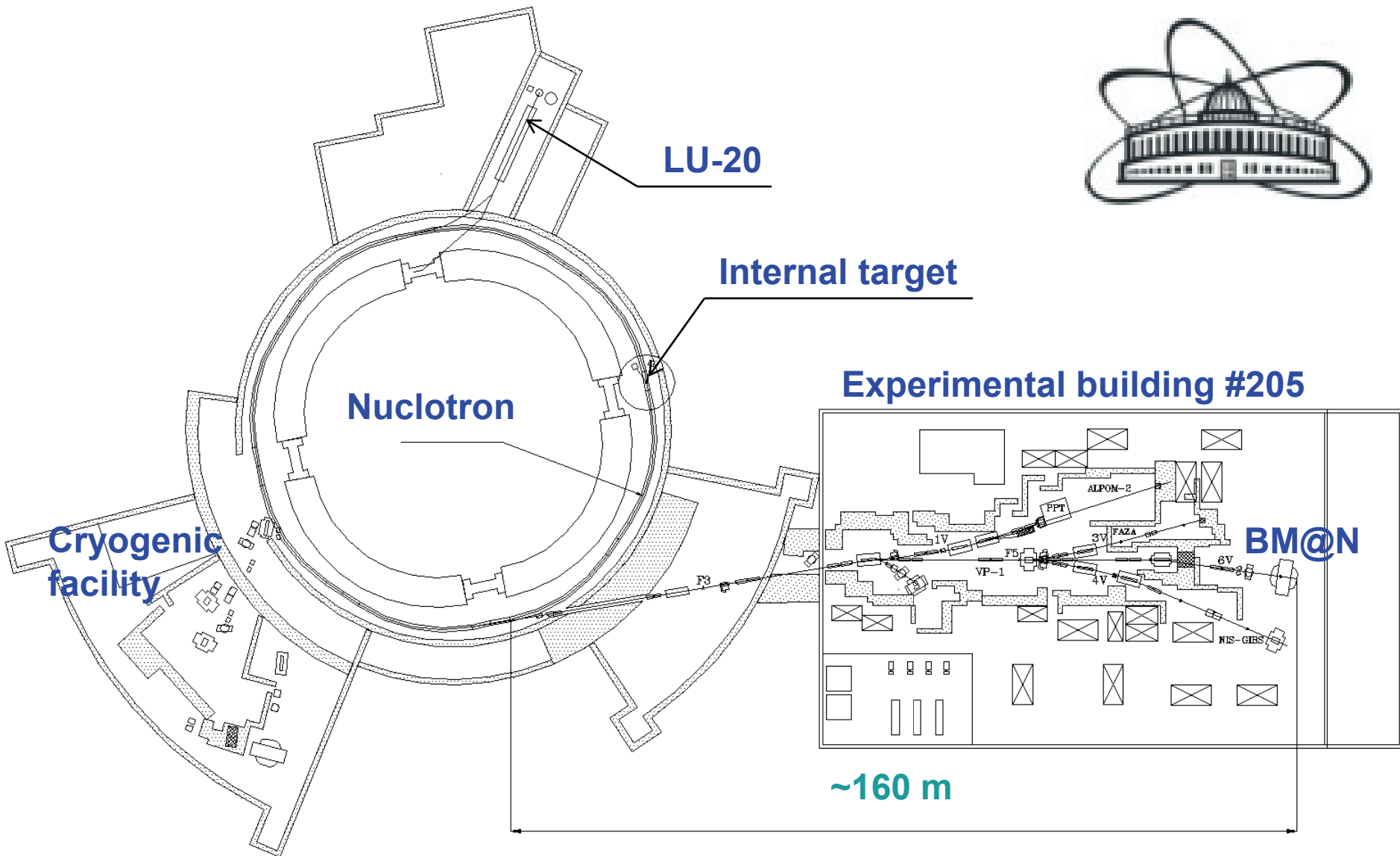
on behalf of Accelerator Division

VBLHEP JINR



RuPAC XXVI, October 2018

VBLHEP accelerator complex





Contents

- **Statistics of the operation in 2017 - 2018**
- **Adiabatic capture**
- **Polarized proton acceleration**
- **Preparation to heavy ion run**
- **Acceleration and slow extraction of low intensity beams**



Statistics of operation

Run #53 26.10-25.12.2016 (1400 h)

SPI optimization, polarimetry

Spin physics experiments

Test of BM@N and MPD elements

Test of the Booster power supply prototypes
with beam acceleration

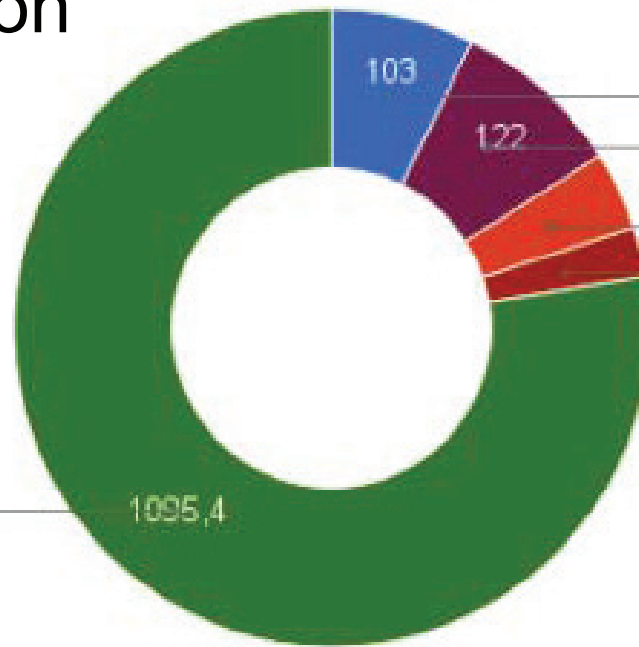
Stochastic cooling, diagnostics, adiabatic capture

Run #53

Time distribution

Machine
development
NICA R&D (24%),
Experiments

77.6%



Cooling, 7.3%

Tuning, 8.7%

Repairs, 4%

Unexpected losses, 2.5%

Polarized, unpolarized deuterons, maximum energy of extracted beam 4.6 GeV/u

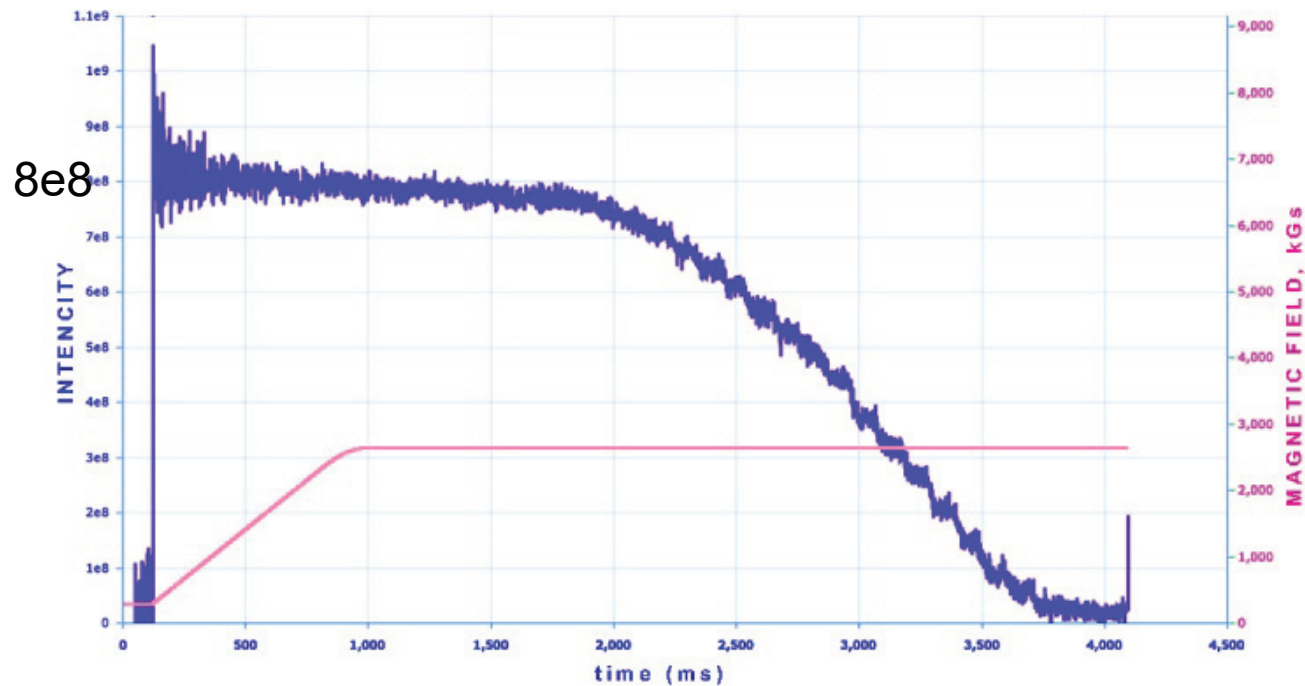


Run #53

Routine operation of SPI and new fore-injector

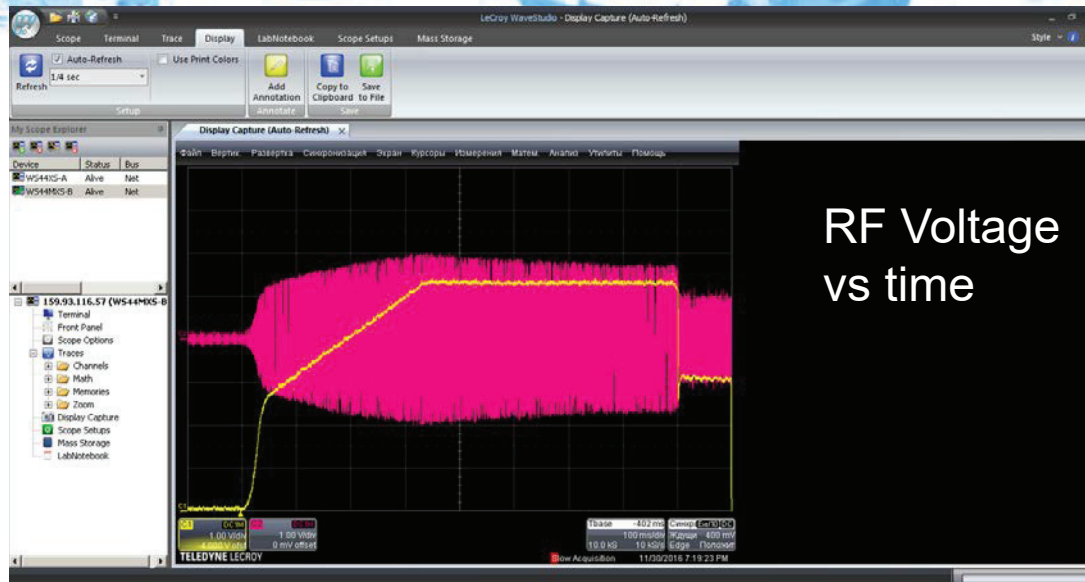
Polarized deuteron acceleration:

Intensity $2\div 5 \cdot 10^8$

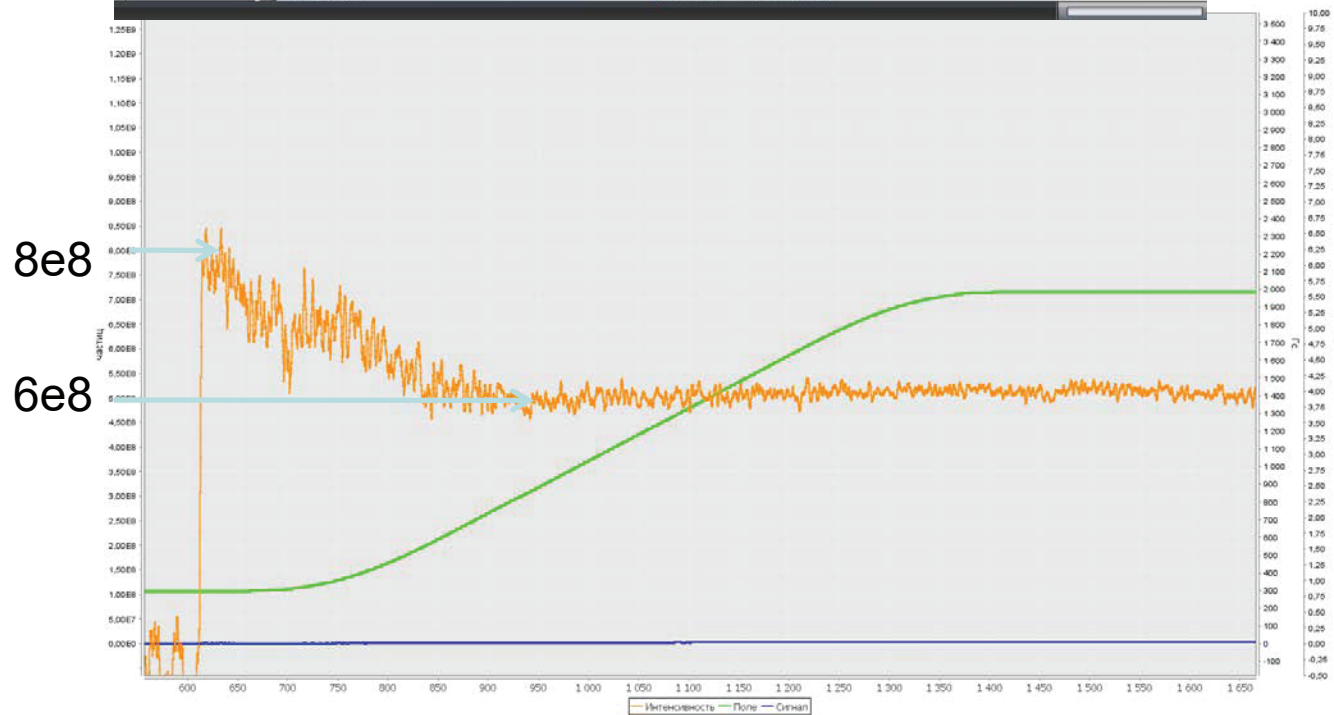


Deuteron Spin Structure experiment, internal target

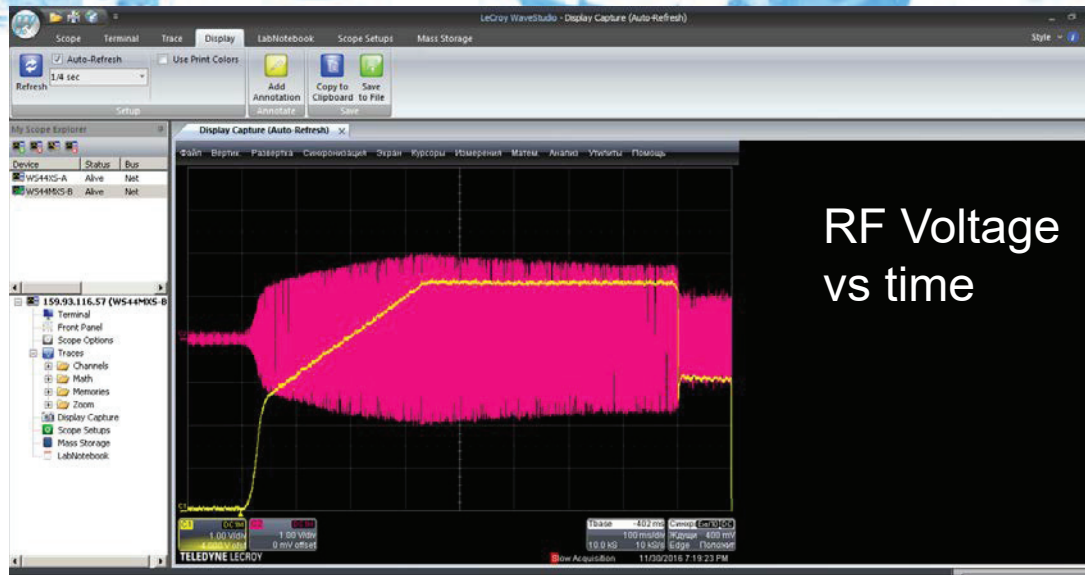
Adiabatic capture



Capture efficiency ~ 70%



Adiabatic capture

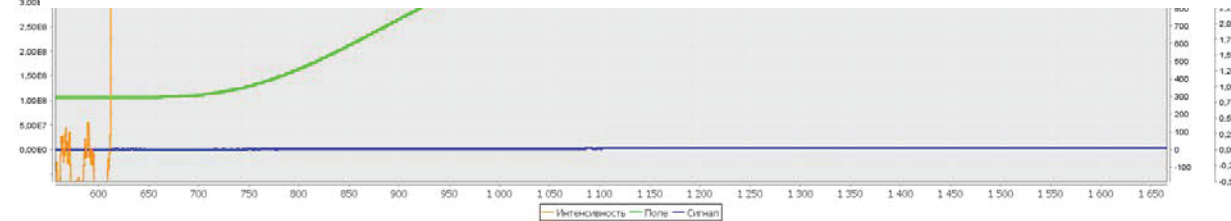


Capture efficiency ~ 70%

**Routine operation during a few shifts
with injection at magnetic field plateau
+ adiabatic capture**

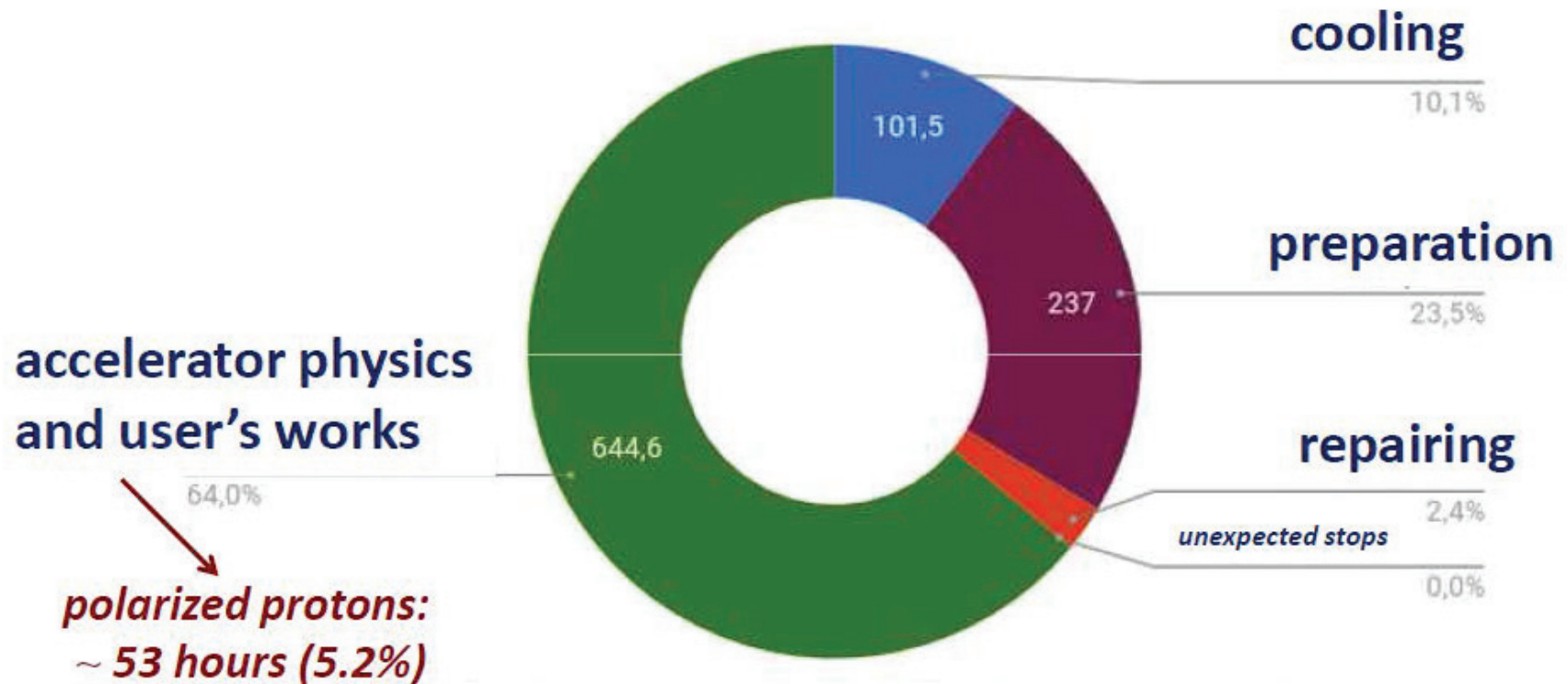
Routine operation during Runs #54, #55

8e8
6e8



Run #54 10.02 - 24.03.2017

The total run duration: ≈ 1008 hours



- Acceleration of polarized and unpolarized deuterons and protons from SPI
- Acceleration of carbon and lithium ions from laser source

Maximum extracted beam energy 5.2 GeV/u (1.85 T)



Proton acceleration

LU-20 was designed for the proton acceleration to 20 MeV
Injection energy of about 625 keV was provided by HV transformer

The deuterons and light ions were accelerated at second harmonics
Injection energy – 156 keV, output energy 5 MeV/u

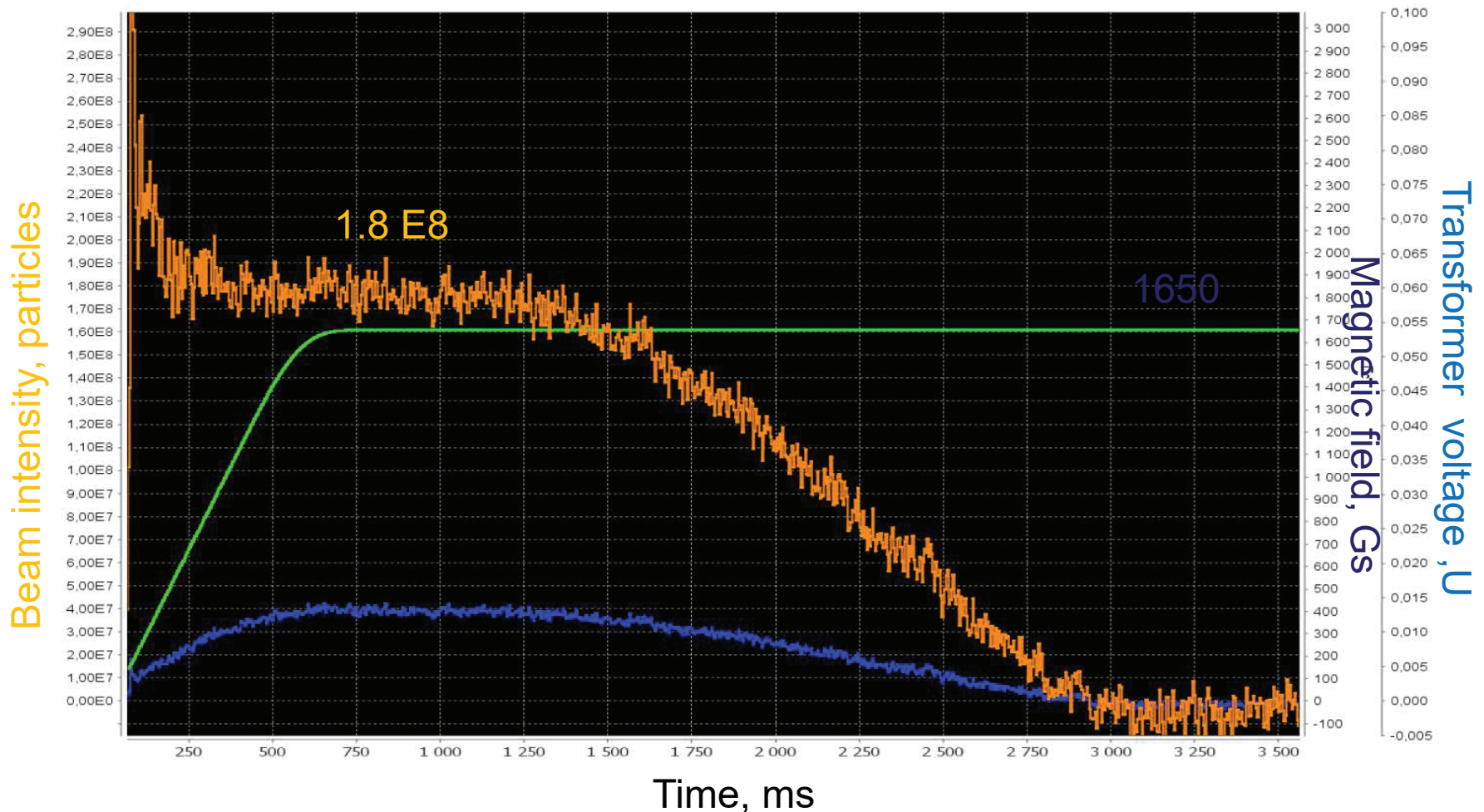
New RFQ fore-injector provides output energy of 156 keV for all ions:

- the protons can be accelerated in LU-20 at the second harmonics only
- the output proton energy is **5 MeV**
- the Nuclotron magnetic field at injection has to be about **150 G**
(instead of 300 G for other ions)

Because of high importance of the proton program 4 shifts were spent for the tuning of the accelerator complex in the new regime:

- Tuning of LU-20 and injection channel
- Orbit correction at injection
- Working point adjustment
- Tuning of acceleration

Proton acceleration



For acceleration of polarized proton beam a spin-rotator in the LU-20 - Nuclotron transfer channel is necessary



Slow extraction

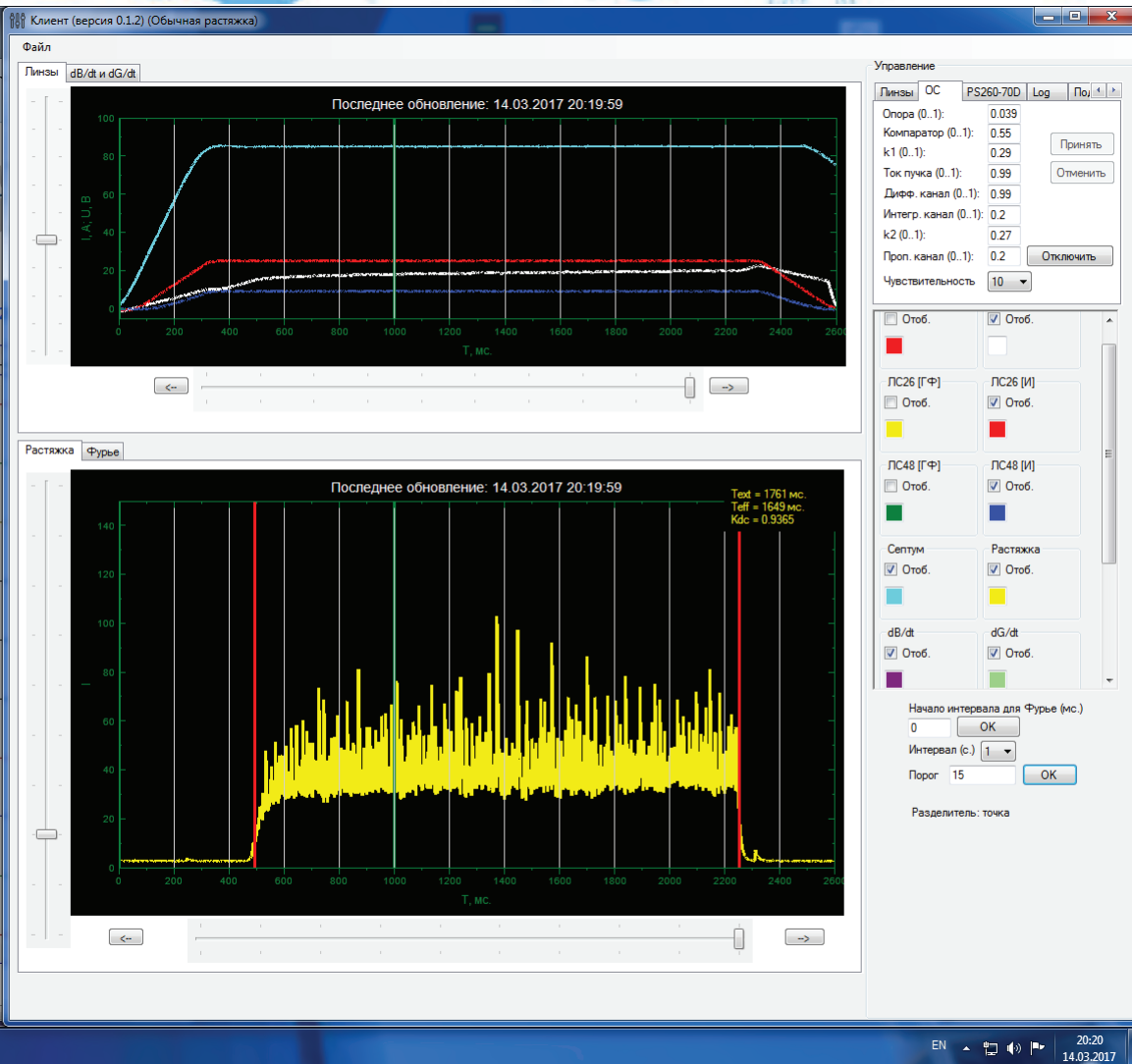
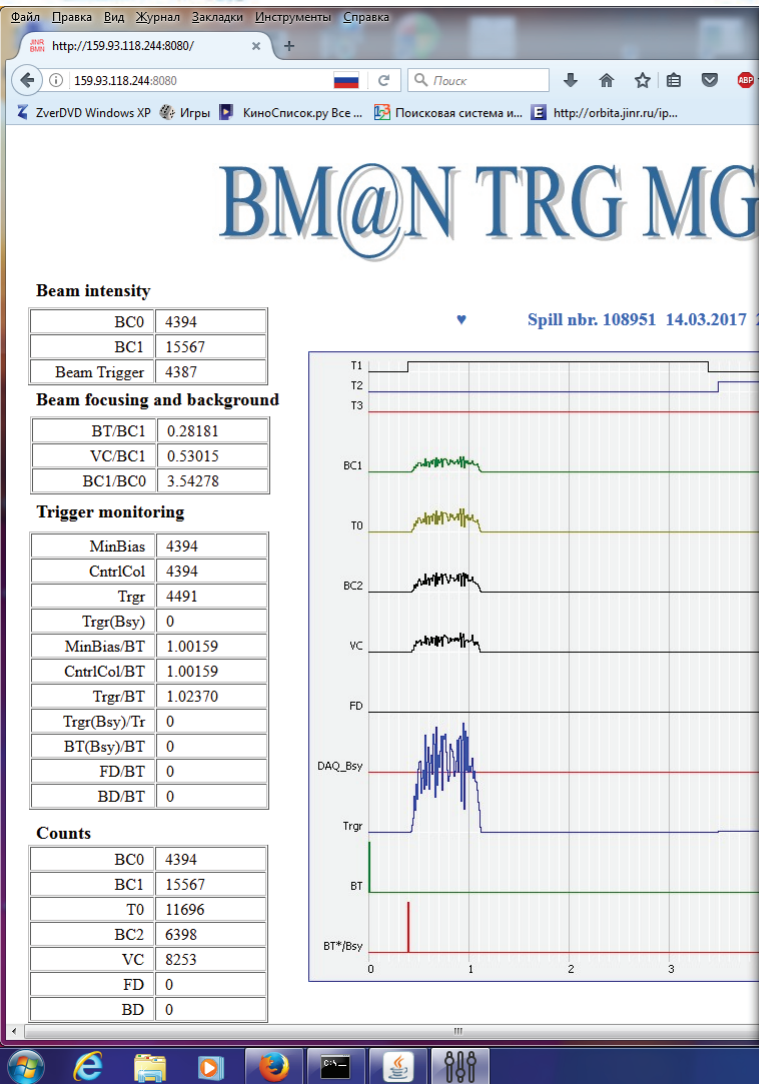
During the run #47 (March 2013) at the beam extraction at 5 GeV/u insulator of Electrostatic septum of the slow extraction system was broken due to HV discharge

The first runs after installation of new insulator the Voltage was limited at the level of about **120 kV** (the good efficiency of the extraction was provided up to energy of about 3 GeV/u only)

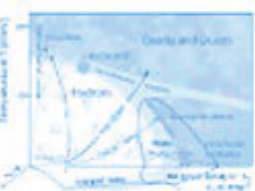
From June 2016 to March 2017 **three** Nuclotron runs were performed at total duration more than **3000 h**

Intensive training of the Electrostatic Septum permitted to increase Voltage up to **150 kV** (for the beam extraction at maximum design energy one needs 160 kV)

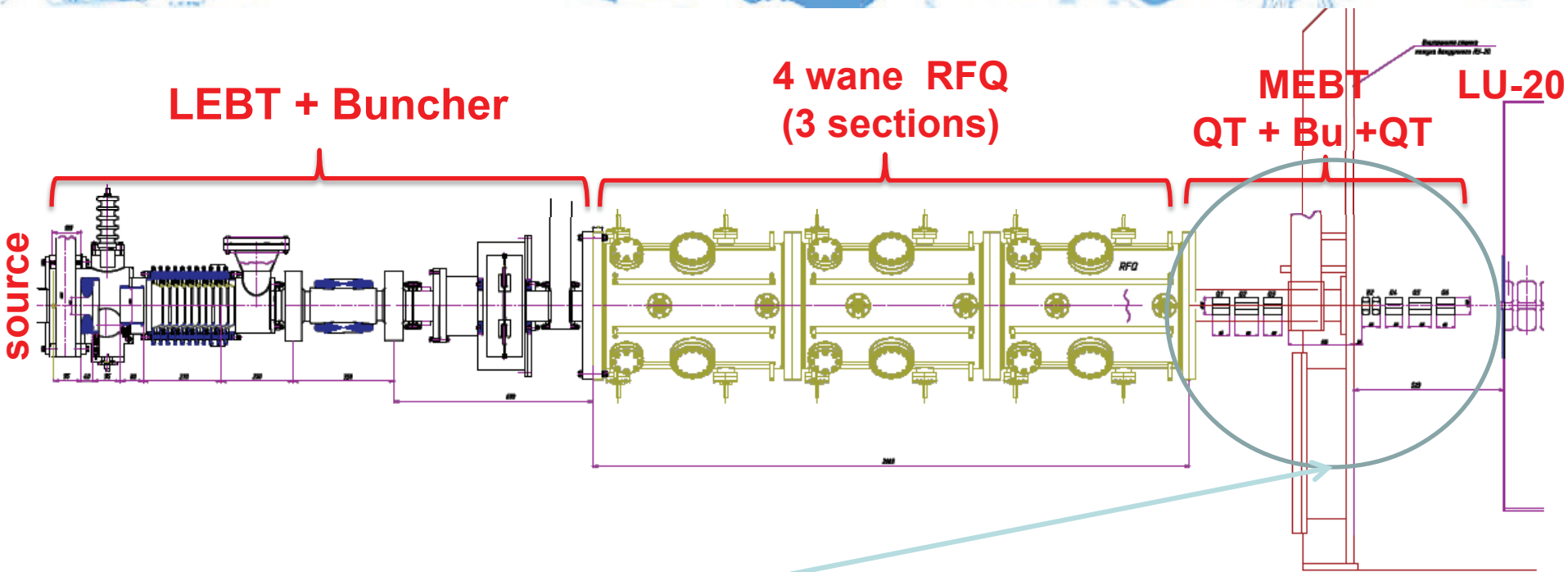
Slow extraction



Run #54, carbon ions, 5.2 GeV/u, $V_{ESS} = 145$ kV



Preparation for BM@N



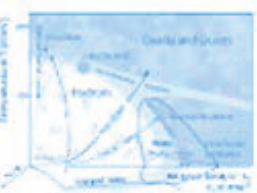
The MEBT includes two triplets of quadrupole lenses and Buncher

At the first stage the Buncher was absent

(Theory transmission ~ 20%, achieved up to 10%)

The buncher is prepared to the run #55:

Expected gain in transmission is about 3 times



Preparation for heavy ion run

Tuning of KRION at test bench

E.D.Donets, E.E.Donets

From April to October

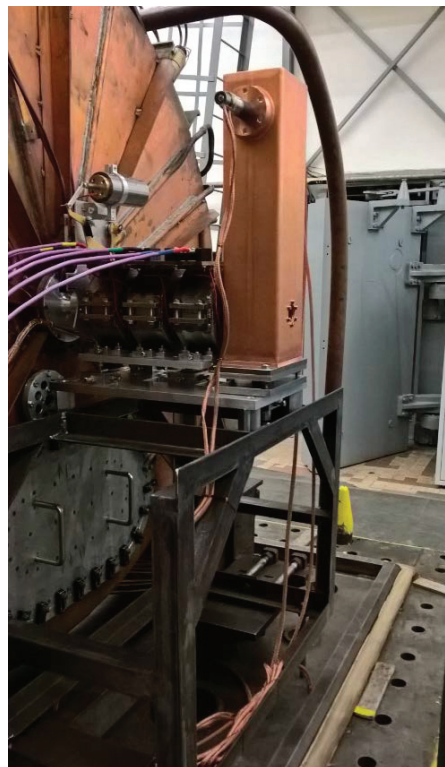
Optimization of the string formation
for generation of Ar^{16+} and Kr^{26+} ions

From October –
start of tuning at LU-20



Buncher tuning and installation (April 2017)

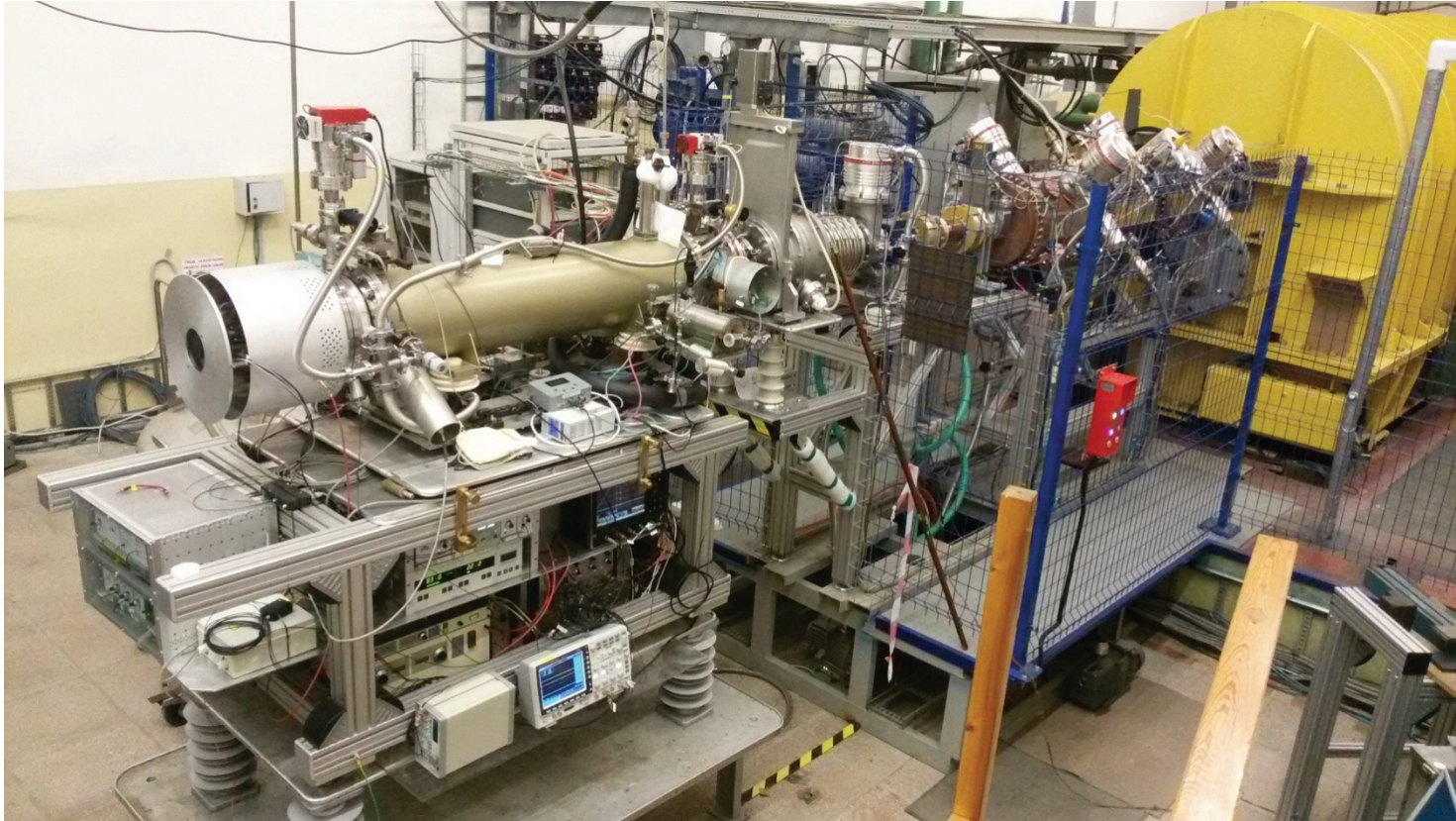
ITEP,
Chernogolovka



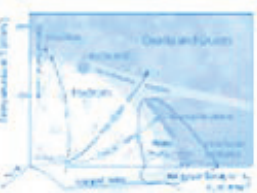
From Jun to September
LU-20 run with the laser source –
tuning of the injection chain

Preparation for heavy ion run

Tuning of the injector complex



Usage of the buncher permitted to increase the Ar beam intensity at the entrance of the Nuclotron by about 5 times



Preparation for heavy ion run

Tuning of the injector complex

Results from 20.12.2017

	Ar 14+	Ar 15+	Ar 16+	C 4+	C 5+	C 6+	Kr 26+
nQ	0.024	0.07	0.052	0.034	0.064	0.18	0.04
Ne	1.5 E8	4.4 E8	3.3 E8	2.1 E8	4.0 E8	1.1 E9	2.5 E8
Nparticle	1.0 E7	3.0 E7	2.0 E7 (3.7 E7 20.01)	5.3 E7	8.0 E7	1.7 E8	1.0 E7

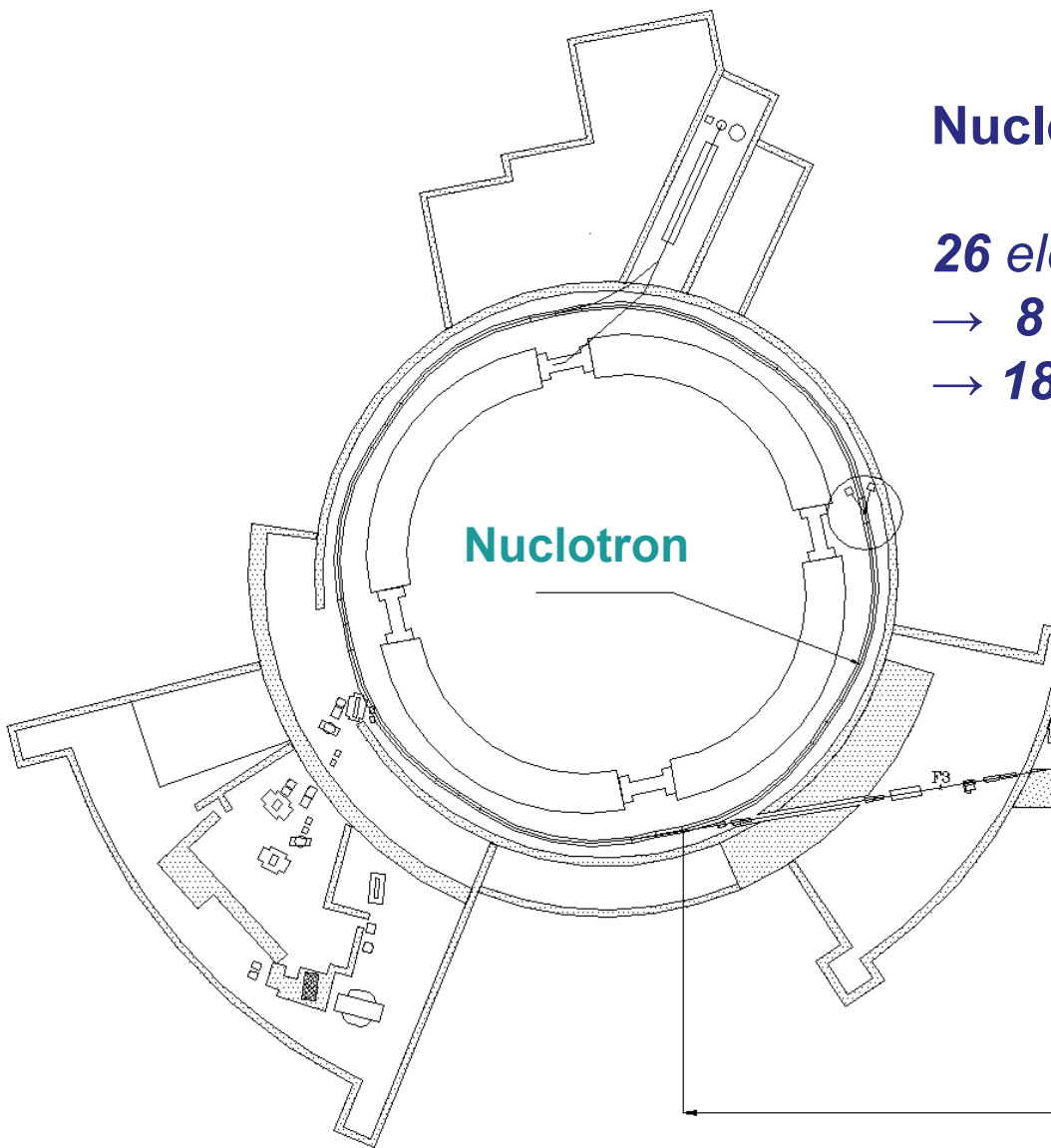
Preparation for BM@N

Nuclotron to BM@N beam line:

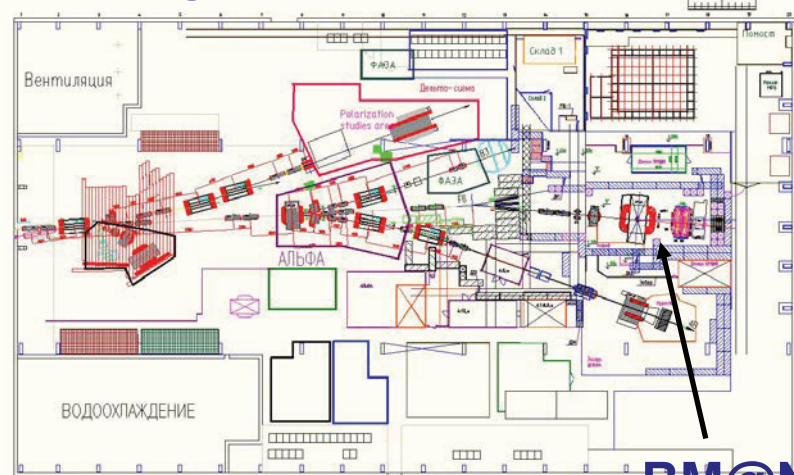
26 elements of magnetic optics:

→ *8 dipole magnets*

→ *18 quadrupole lenses*



Building 205

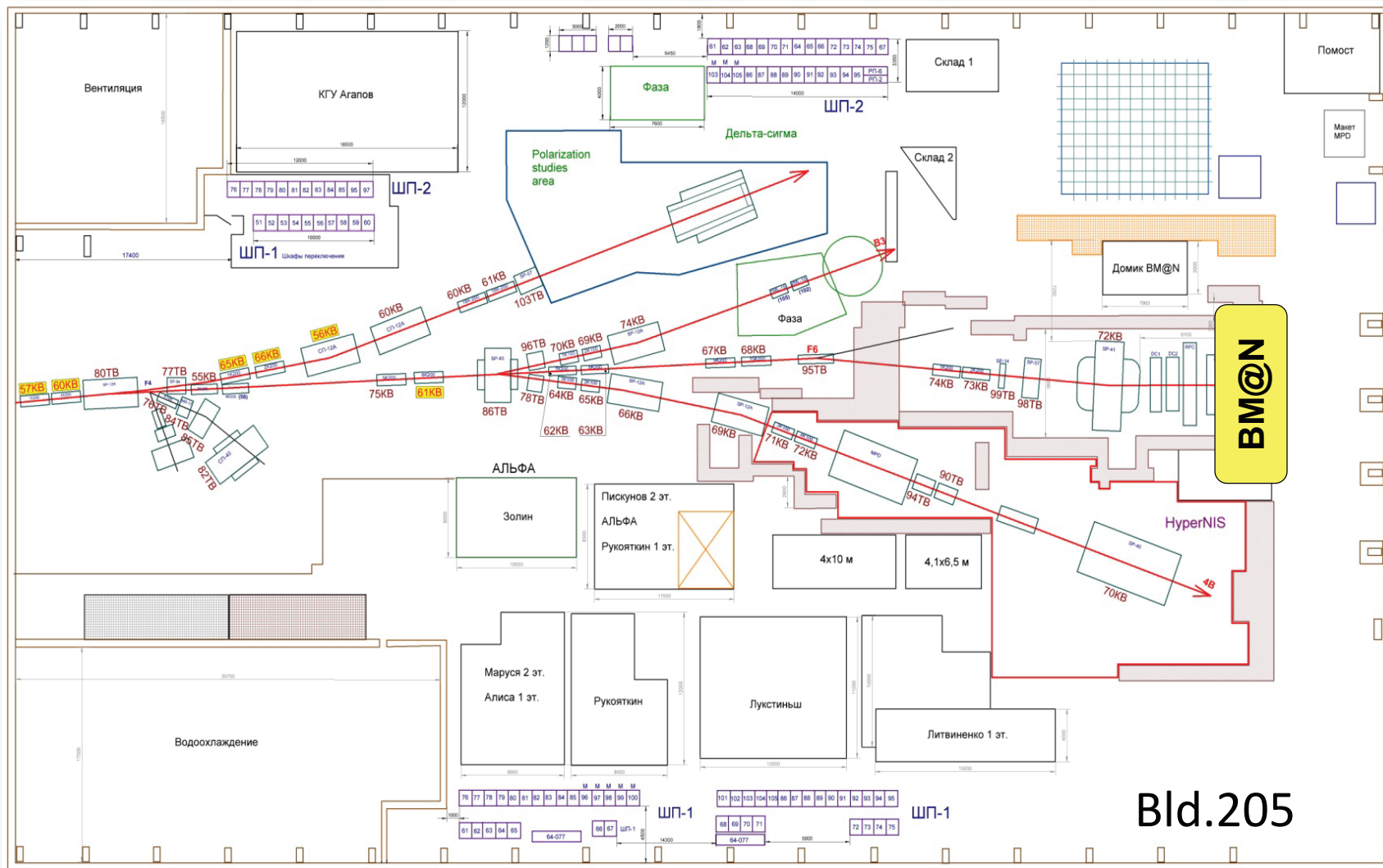


~160 m

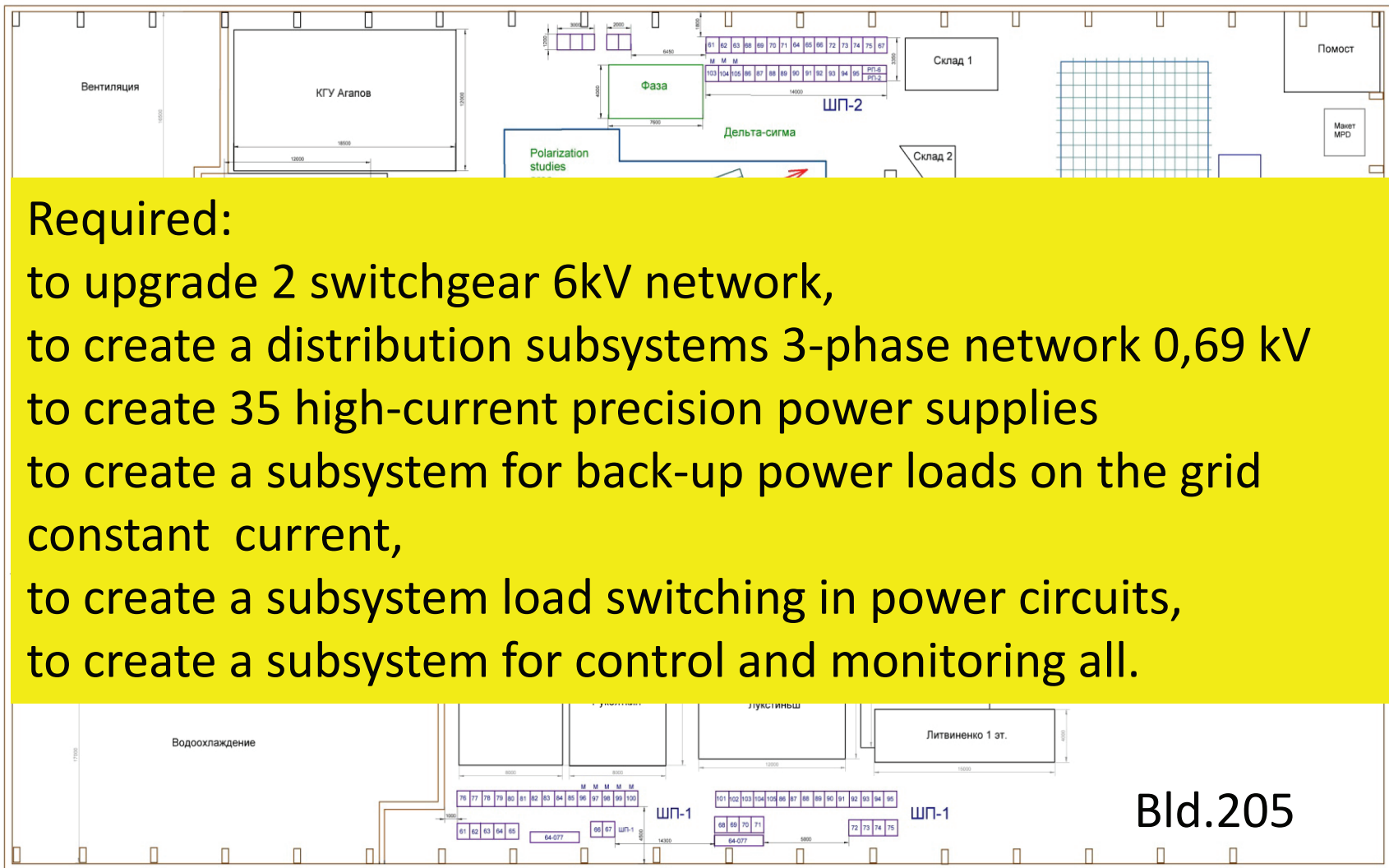
BM@N

Preparation for BM@N

Development of power supply system



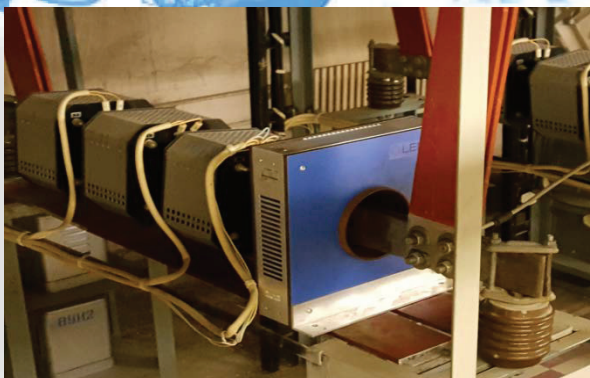
Preparation for BM@N



Creating power source system



Supply unit 300A x 8V BM@N



LEM 5 kA, bld 205



Remote control system



Operated during 2 runs



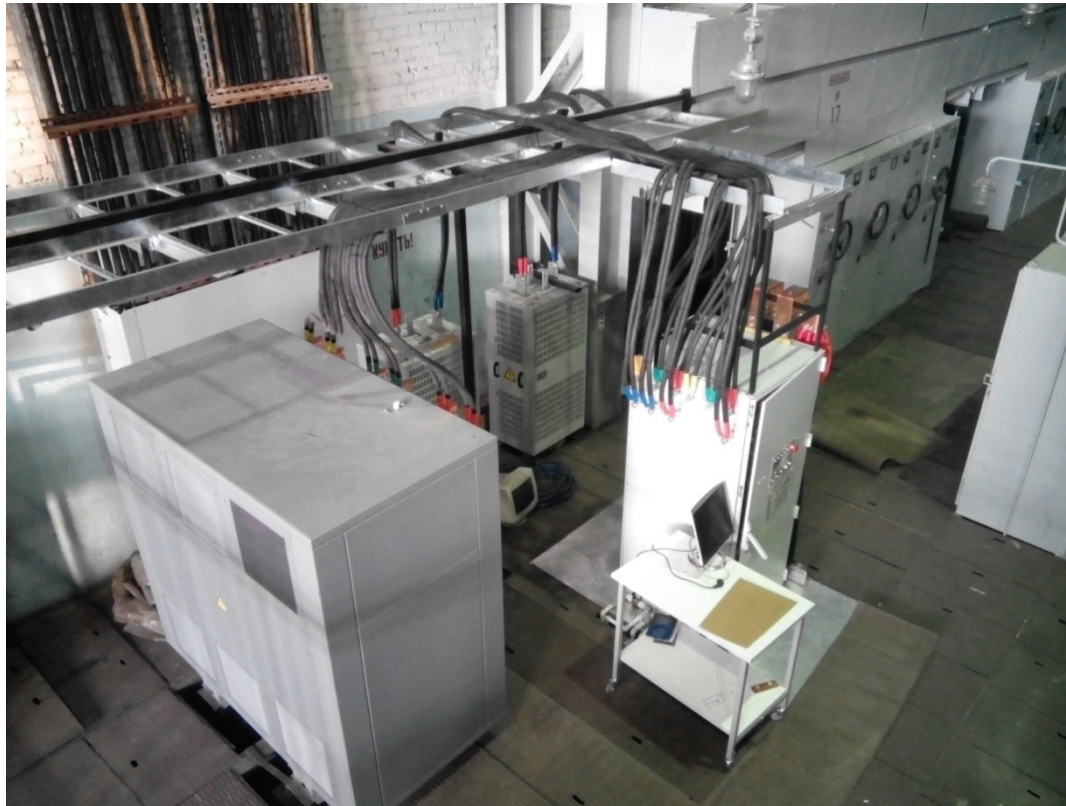
Active filter АФВТ 600-25
ЛМ Инвертор



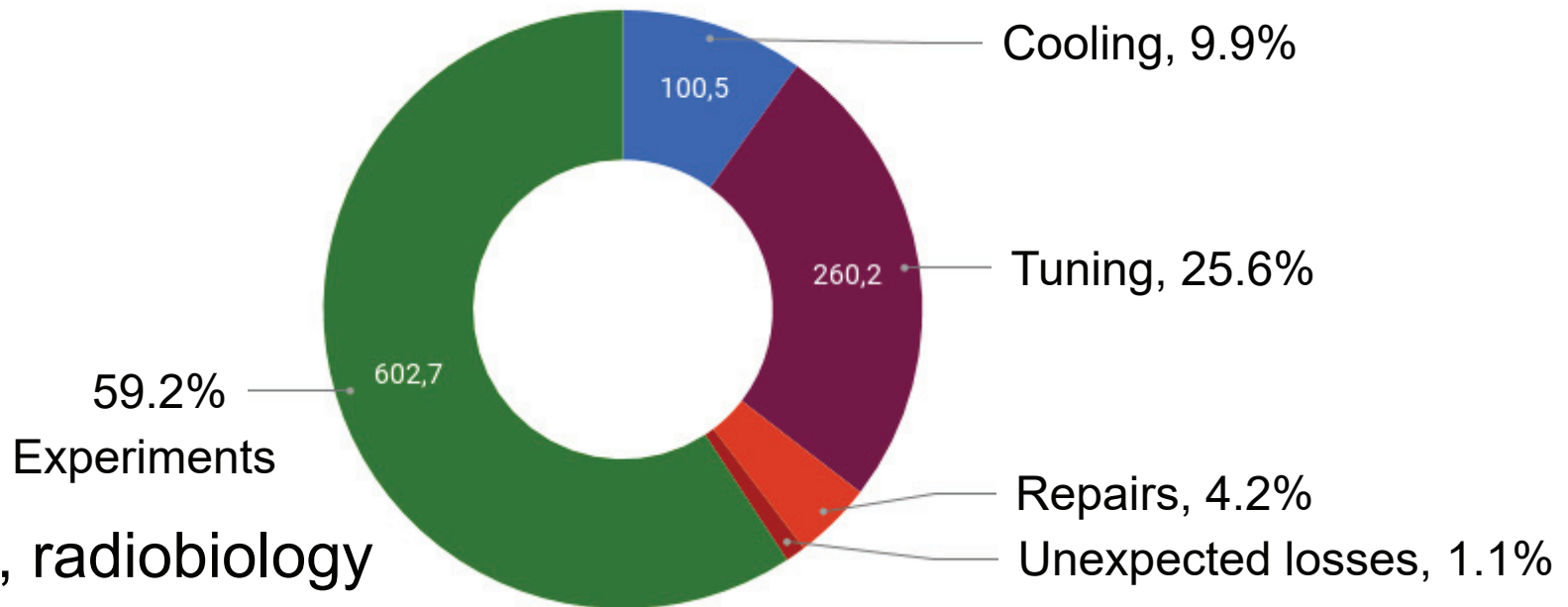
Switchgear ПС13, бд.1А

Creating power source system

Supply unit ИП2500-280 for BM@N large aperture magnet
(relative stability 10^{-4})



Was operated in the run #55



C ($5 \cdot 10^8$), Ar ($2 \cdot 10^6$), Kr ($5 \cdot 10^5$) ions,
magnetic field 1.3 – 1.8 T



Heavy ion acceleration and extraction

- Tuning of the ion source and injection chain
- Adiabatic capture
- Combination of resonant and stochastic extraction
- Tuning of the ring at carbon beam
- Development of low intensity beam diagnostics
- Improvement of vacuum conditions

Tuning of the ion source and injection chain

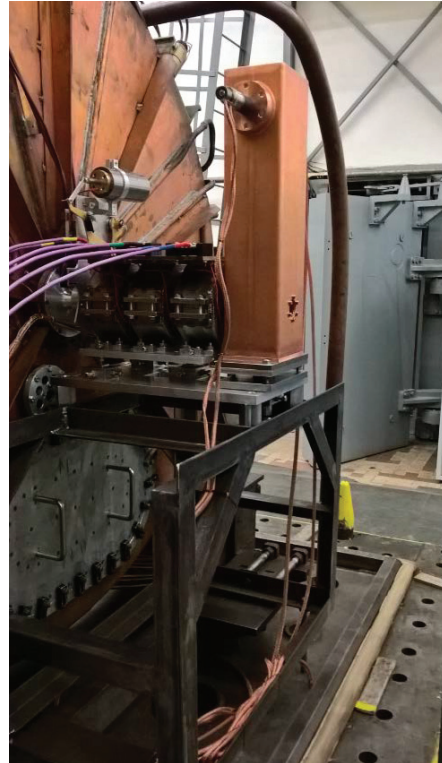
Tuning of KRION at test bench

From April to October
Optimization of the string formation
for generation of Ar^{16+} and Kr^{26+} ions
From October –
start of tuning at LU-20



Buncher tuning and installation (April 2017)

ITEP,
Chernogolovka

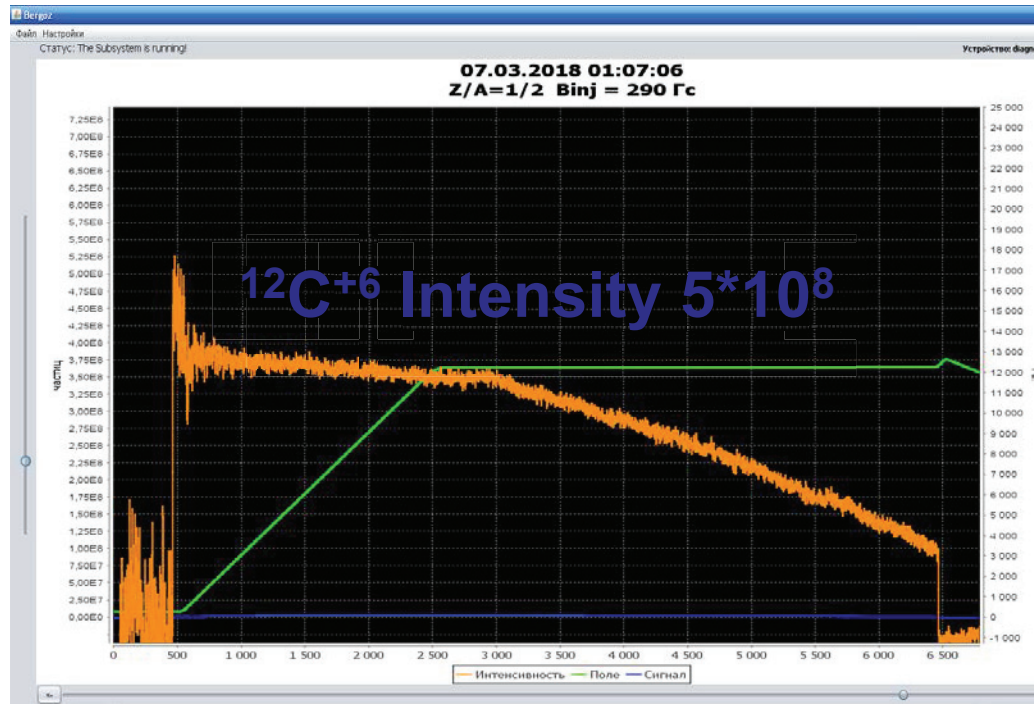


From Jun to September
LU-20 run
with the laser source

**Usage of the buncher permitted
to increase the Ar beam intensity
at the entrance of the Nuclotron by about 5 times**

Tuning of the Nuclotron

$^{12}\text{C}^{+6}$ beam acceleration (3,5 GeV/u)



Beam injection, adiabatic capture, orbit correction, beam acceleration were adjusted at high intensity using standard diagnostics

Beam diagnostics at low intensity

Diagnostics at slow extraction

Diagnostics in experimental hall

Ionization profile monitor

Вывод пучка из
Нуклофона в корпус 205

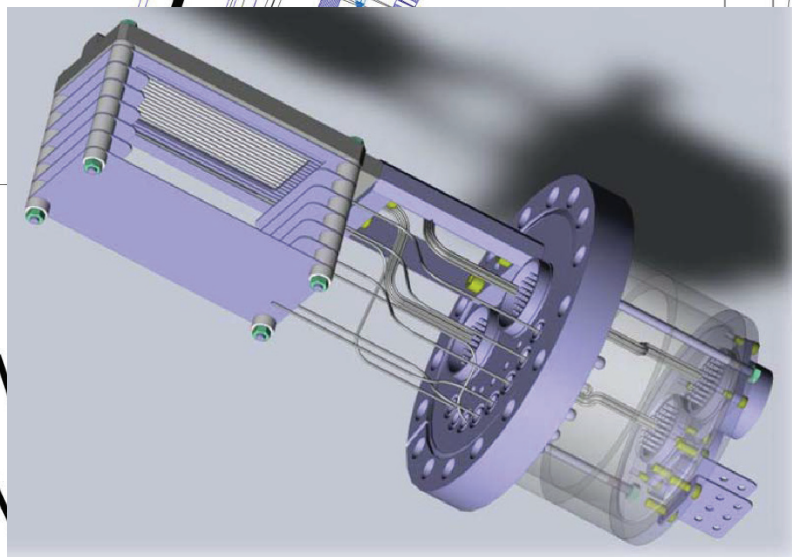
Вывод пучка из
Бустера в Нуклотрон

Этажа
обслуживания 30

Пучок из
НТЛАС

Пучок из РН20
Нуклофона

Beam diagnostics at low intensity



**Monitor on MCP
from 26.02 till 5.04:
on-line information
in Nuclotron and LU-20
control rooms**

**Ionization profile
monitor**

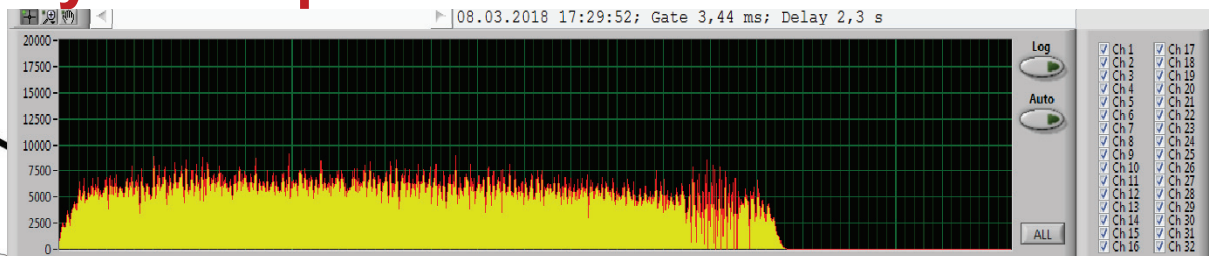
Beam diagnostics at low intensity

Diagnostics
at slow
extraction



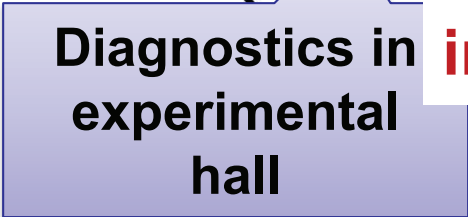
MCP monitor (test operation)

Scintillation horoscope:
Dynamic profiles of the extracted beam



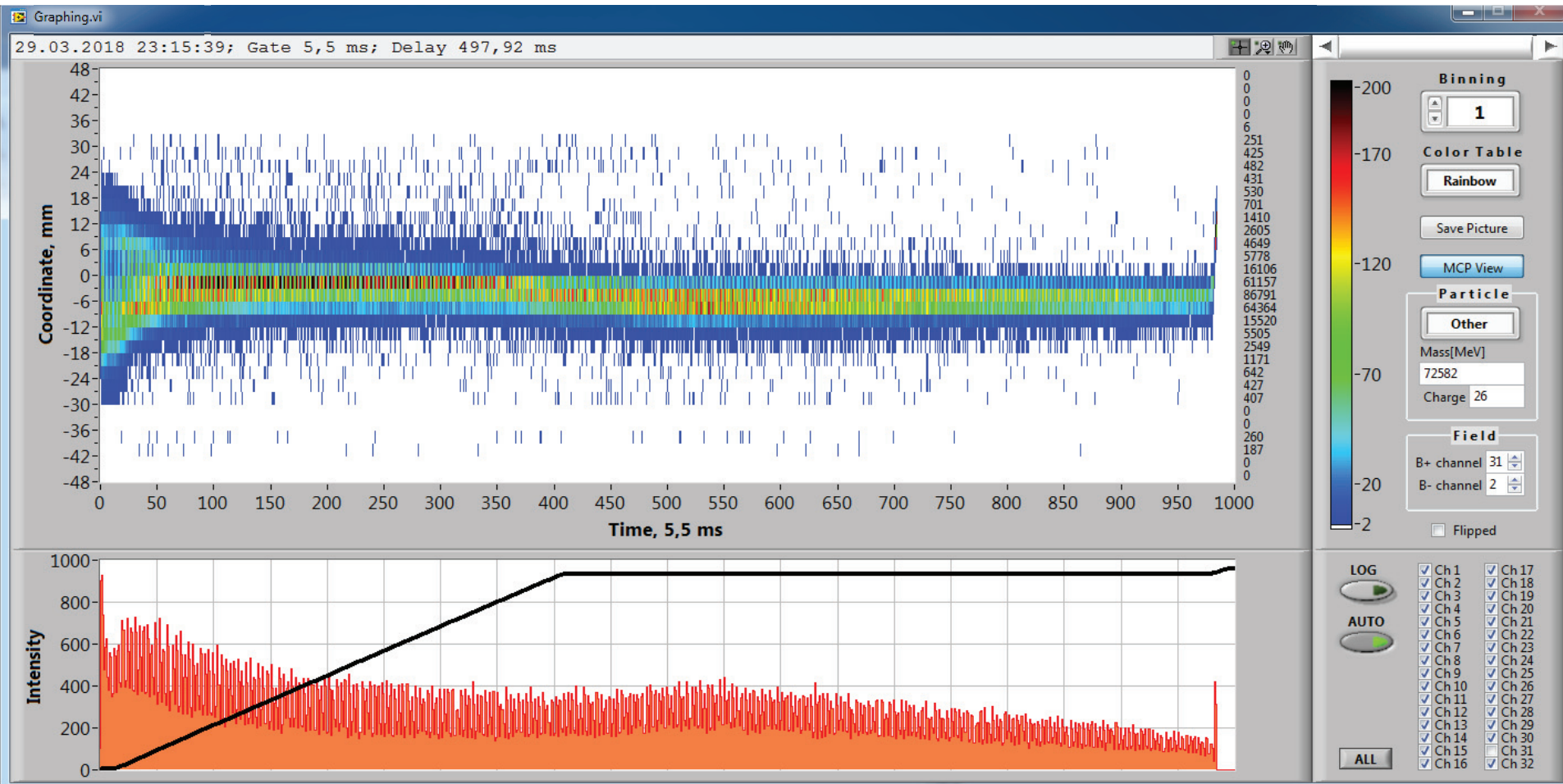


Beam diagnostics at low intensity



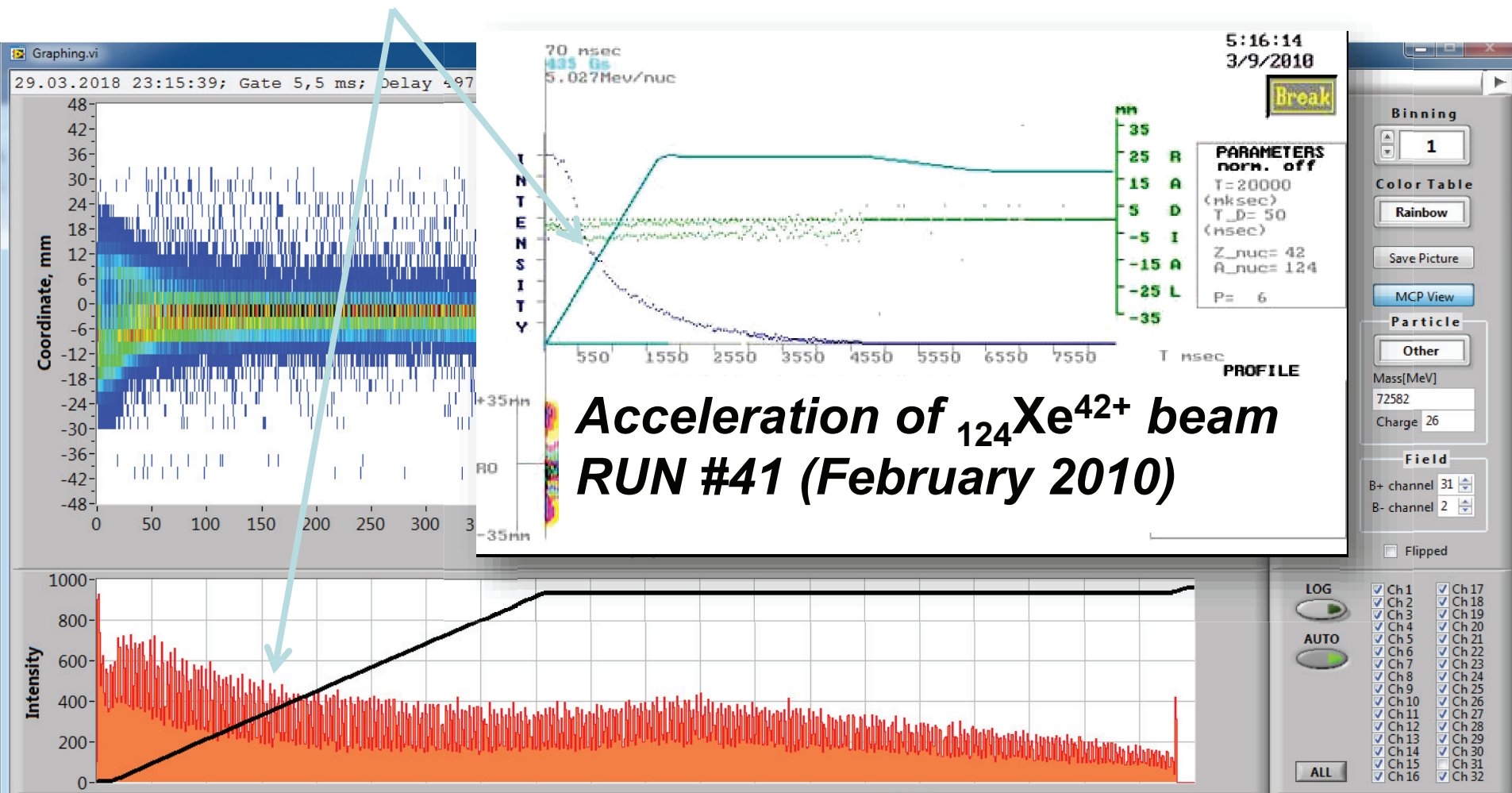
Beam diagnostics at low intensity

$^{78}\text{Kr}^{+26}$ beam acceleration (3,2 GeV/u) **RUN #55**

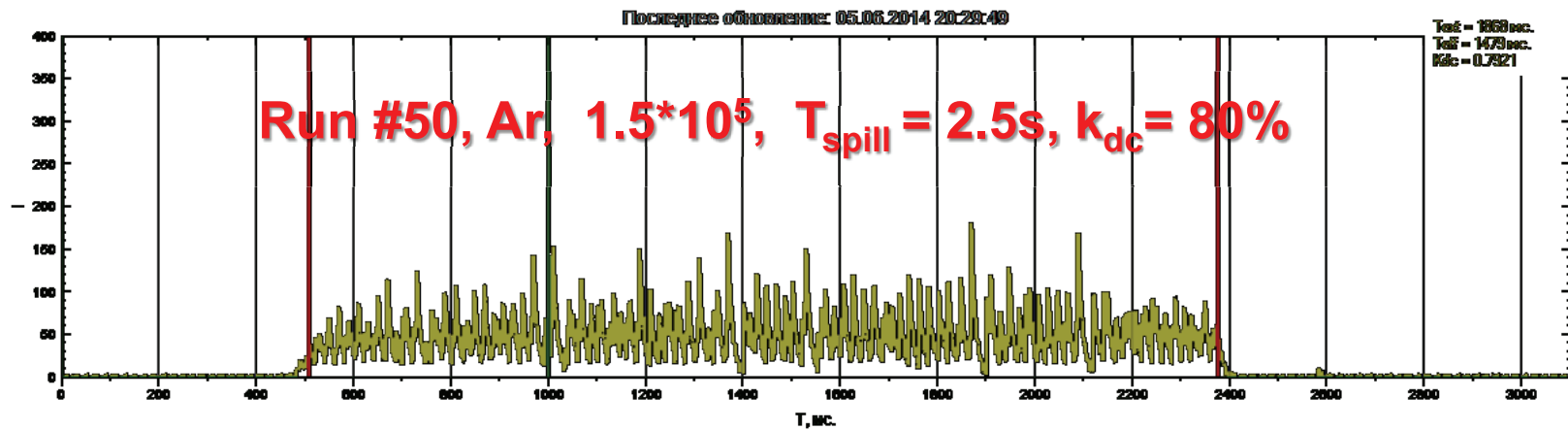
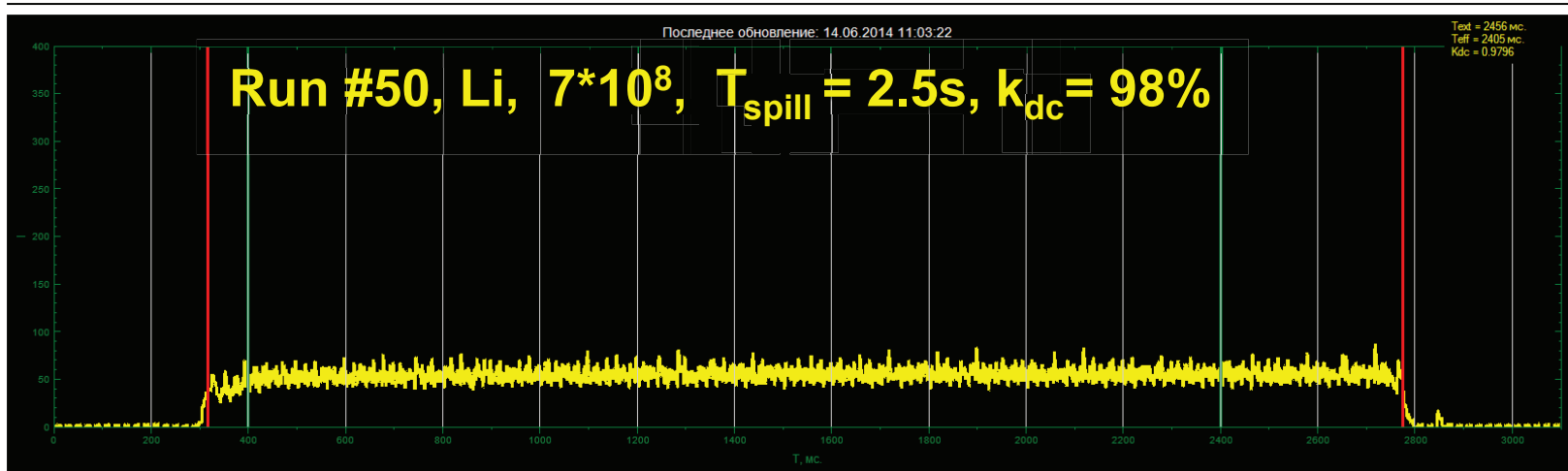


Improvement of vacuum conditions

Losses on residual gas



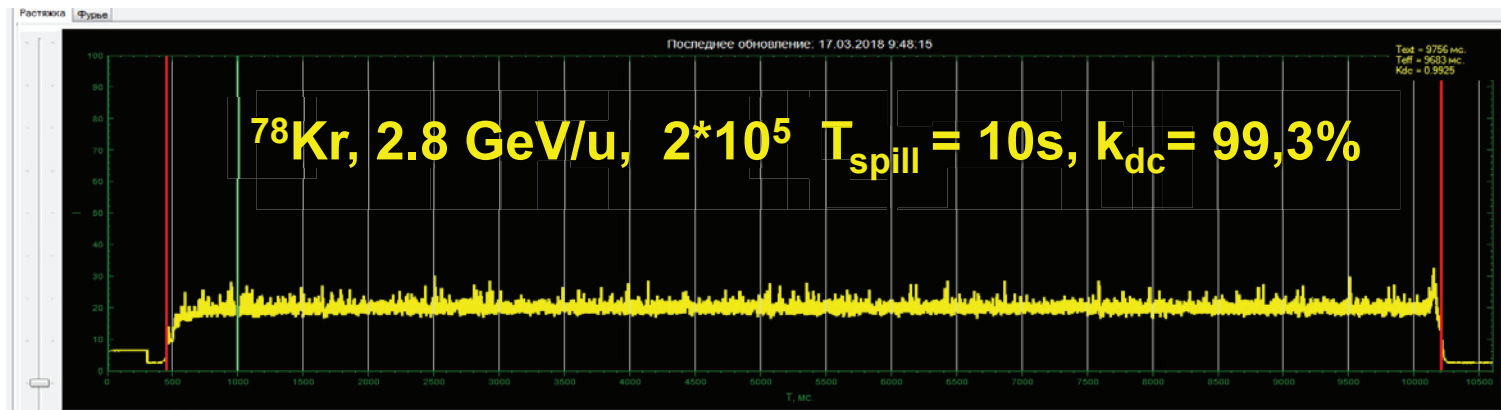
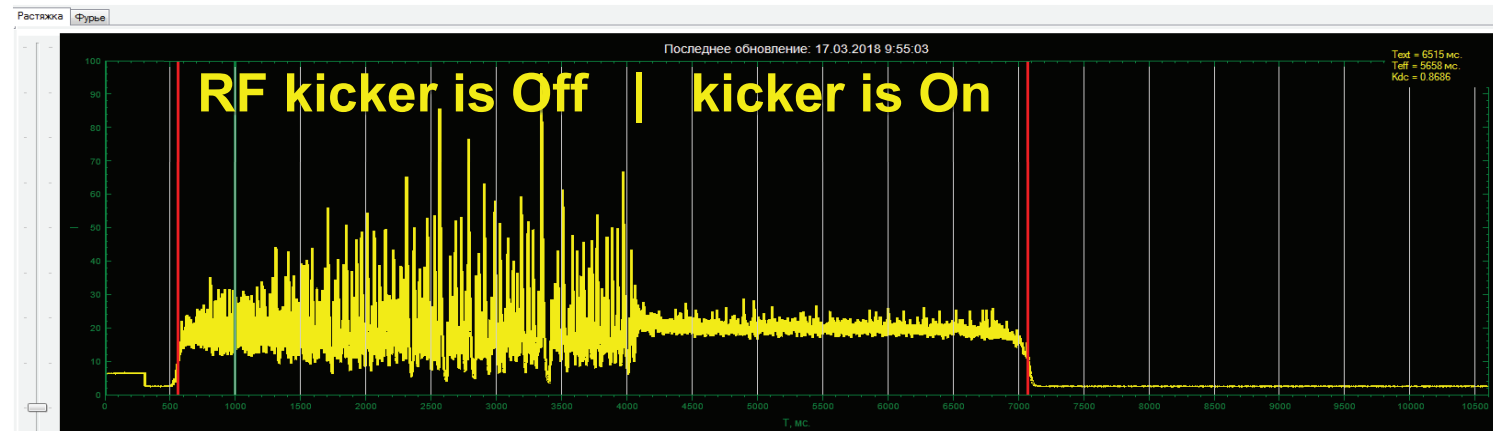
Slow extraction at low intensity



Slow extraction development

Run #54 – test of uncontrolled stochastic slow extraction (RF knock-out) using diagnostic kicker of Q-meter.

Run #55 – routine operation of combined method: controlled displacement of working point into 1/3 resonance + RF knock-out

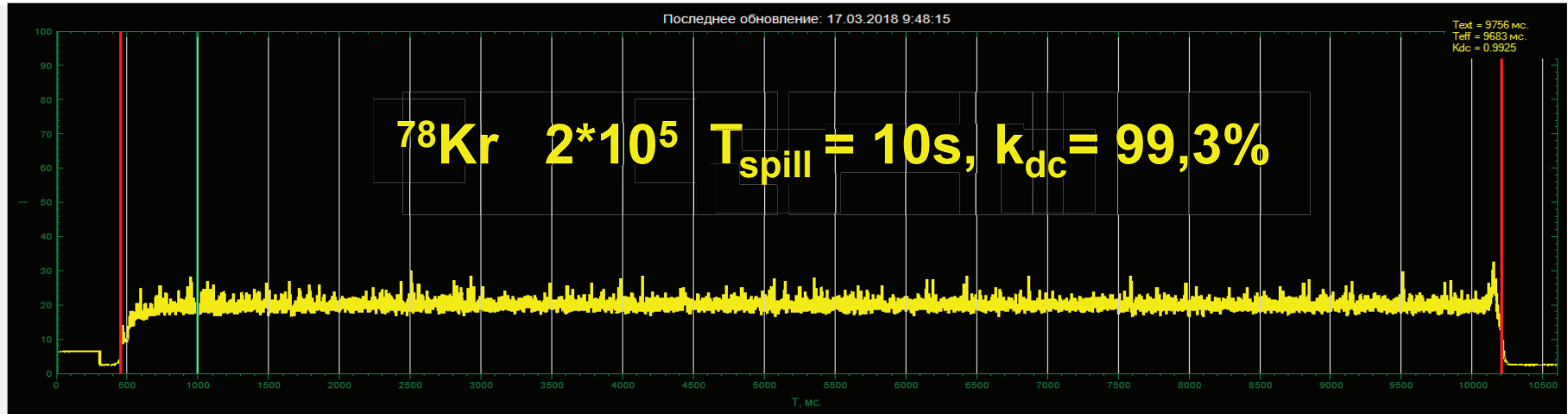
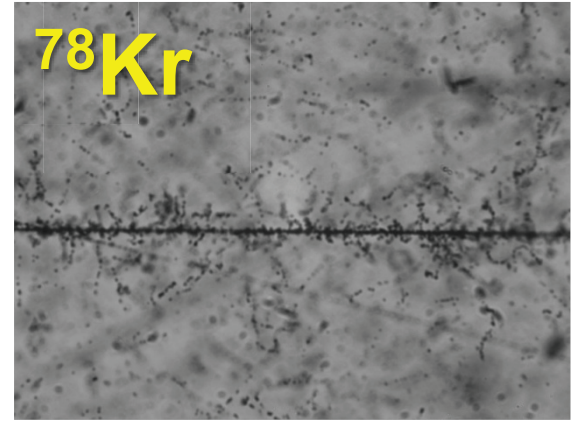
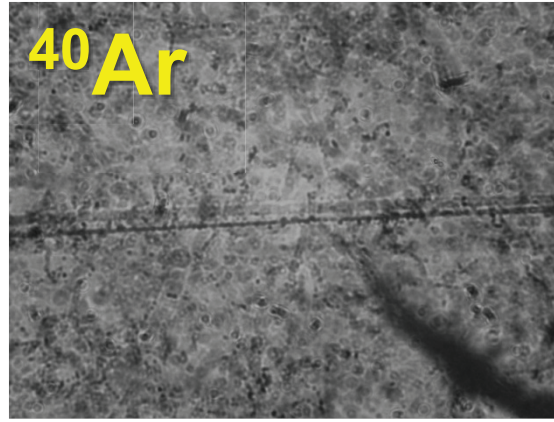
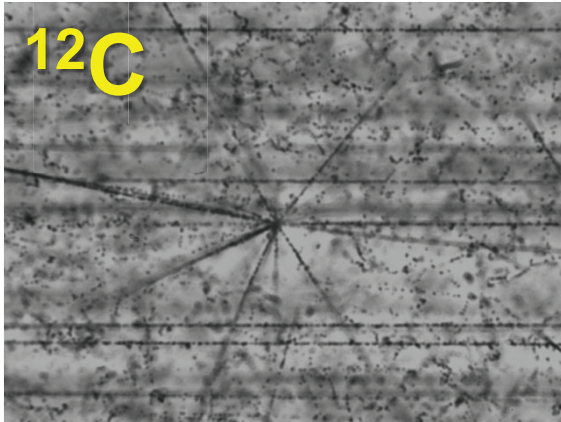




Conclusions

- Adiabatic capture is used in routine operation
Capture efficiency $\sim 70\%$
- Slow extraction system provides required beam quality
At Magnetic field up to 1.85 T
At the intensity range from 10^4 to 10^{10} particles/sec
- Possibility of the polarized proton acceleration
was demonstrated
- Assembly of the NICA Booster has been started.
Technological run is scheduled for the end of 2018

NICA experimental program has been started



Thank you for attention